

Jeffrey Di Perna

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EDUCATION

Queen's University

Kingston, ON

Bachelor of Applied Science — Major in Computer Engineering

Expected May 2028

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Data Science

EXPERIENCE

AI Co-op

May 2026 – Aug 2026

4AG Robotics

Salmon Arm, BC

- Trained an **object detection and classification model** on approximately **2,000 images** using **transfer learning**, achieving **97%+ classification accuracy** for a robotic mushroom-harvesting system
- Enhanced an internal annotation tool using **Python** and **Qt**, increasing **labeling speed by 31%** and accelerating **supervised learning dataset preparation**
- Managed and quality-checked annotated datasets using **Roboflow**, improving **label consistency** and reducing **storage credit usage by 17%** through data migration to **local hard drives**
- Prepared and analyzed **real-mushroom measurement data** to create **representative test objects**, enabling **realistic system performance testing**

Machine Learning Engineer

Sep 2024 – April 2026

Queen's Machine Intelligence and Neuroevolution Design Club (QMIND)

Kingston, ON

- Developed RAG-driven instruction generation using **Python**, piloted with **100+ patients** in **real clinical use**
- Enabled self-serve clinician uploads and **QR code** delivery to improve information for post-surgical operation care
- Synthesized generative AI responses with scientific literature and patient medical data to ensure optimal patient guidance, achieving **89% accuracy with physician recommendations**

Software Engineer

Sep 2025 – April 2026

Engineering Society Software Development Team

Kingston, ON

- Created a service that analyzes resume structure and keywords, **improving interview callback rates by 30%**
- Leveraged **NLP** to analyze resumes and enhance effectiveness for **ATS systems** on a specific job posting
- Reduced resume editing time by **35 minutes** per user and integrated the project into Queen's official career services

Systems Administrator

Jun 2025 – Aug 2025

Trillium Health Partners

Mississauga, ON

- Built virtual data center models in **Sunbird DCIM** to monitor **20+** server racks in real time across multiple locations
- Executed hardware install/upgrade/decommission cycles to scale data centers for patient medical record storage
- Integrated **environment sensors for temperature and voltage** to improve response time for data center issues by **18%**

PROJECTS

AI ASL Recognition System | *Python, CNNs, Supervised Learning* | [GitHub](#)

Sep 2024 – Mar 2025

- Engineered a deep-learning ASL gesture detection tool trained on **2000+ videos**, achieving **90% accuracy**
- Adapted 3D→2D CNN with temporal stacking, **cutting required compute by 62%**
- **First place** at Microsoft Firm Day Hackathon
- **Winner of the Social Impact Award** at the Canadian Undergraduate Conference for Artificial Intelligence

AI Stock Price Predictor | *Python, PyTorch, LSTM* | [GitHub](#)

Dec 2025 – Jan 2026

- Developed a time-series forecasting pipeline that processes historical stock data using **Python, PyTorch**, and a **long-short term memory (LSTM) architecture**
- Trained the model to achieve an **error < 7%** relative to actual stock price using **Adam optimization**

SKILLS

Languages: Python, C/C++, Java, JavaScript, HTML/CSS, Assembly, MATLAB, Verilog/VHDL

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Hugging Face, Mongooose

Frameworks: React.js, Next.js, Flask, Qt

Developer Tools: Git, Docker, Linux, MongoDB, REST APIs, SQL, Jupyter, Roboflow, CUDA